Material Behavior Prediction

Milestone 4 Baseline Trained

Complete Technical Guide to Milestone 4: Baseline Model Training & Interpretation

1. Why Always Start with a Baseline Model?

A frequent mistake made by machine learning beginners is jumping straight to complex black-box algorithms like Random Forests, XGBoost, or Neural Networks.

The Scientific Purpose of a Baseline

1. **Sanity Check:** Proves that the data pipeline, preprocessing, and evaluation code work end-to-end without bugs.
2. **Benchmark for Justification:** More complex models require more compute, have higher latency, and are harder to interpret. To justify using a complex model, it *must* beat the simple baseline by a meaningful margin.
3. **Direct Interpretability:** Linear regression provides explicit mathematical weights for each material component that can be checked against known physical chemistry.

2. The Mathematical Baseline: Ordinary Least Squares (OLS)

Linear Regression models predicted compressive strength ($\hat{y}$) as a weighted linear combination of the 8 standardized features:

$$\hat{y} = 35.07 + 12.12(\text{cement}) + 8.44(\text{slag}) + 6.89(\text{age}) + 5.44(\text{ash}) + 1.97(\text{superplastic}) + 1.67(\text{fineagg}) + 1.20(\text{coarseagg}) - 2.83(\text{water})$$

Standardized Model Weights (Coefficients) & Material Science Meanings

Feature	Standardized Weight	Direction	Physical & Chemical Interpretation
cement	**+12.12 MPa**	Positive (↑)	Primary hydraulic binder forming calcium silicate hydrate (C-S-H) load-bearing matrix.
slag	**+8.44 MPa**	Positive (↑)	Latent hydraulic binder; reacts over time to provide strong structural density.
age	**+6.89 MPa**	Positive (↑)	Hydration kinetics: longer curing in water increases crystal density and compressive strength.
ash	**+5.44 MPa**	Positive (↑)	Pozzolanic reaction consumes calcium hydroxide to form secondary strength-bearing gel.
superplastic	**+1.97 MPa**	Positive (↑)	Improves cement grain dispersion and paste workability.
fineagg / coarseagg	+1.67 / +1.20 MPa	Positive (↑)	Structural skeleton filling volume and resisting deformation.
water	**-2.83 MPa**	Negative (↓)	**Confirms Abrams' Law:** Excess water evaporates, leaving porous capillary voids that decrease compressive strength.

3. Performance Evaluation on Unseen Test Data

We evaluated the baseline model on the 201 unseen test samples across three standard regression metrics:

Metric	Training Set (804 samples)	Testing Set (201 samples)	Interpretation
R^2 Score	**0.6099 (61.0%)**	**0.5802 (58.0%)**	Proportion of variance explained. 58% of strength variation is captured by a simple linear plane.
MAE	**7.96 MPa**	**8.90 MPa**	Average absolute error. On average, predictions deviate by ~8.9 MPa from true crushed strength.
RMSE	**10.00 MPa**	**11.19 MPa**	Root mean squared error. Penalizes occasional large errors more severely than MAE.

Overfitting Check: Train R^2 (0.61) and Test R^2 (0.58) are very close, confirming the model has not memorized the data.

4. Diagnostic Visualizations & Error Analysis

1 07_baseline_actual_vs_predicted.png • Actual vs. Predicted Strength

What to Observe:

- Points cluster reasonably close to the red dashed 45-degree line ($y = x$), showing moderate linear correlation.
- **Deficiency Observed:** For high-strength concrete (>60 MPa), the linear model consistently underpredicts. It flattens out and struggles to predict high strengths because linear regression cannot bend to capture exponential or multiplicative interactions between ingredients.

2 08_baseline_residuals.png • Residual Error Distribution

What to Observe:

- Residuals ($e_i = y_{true} - y_{pred}$) are roughly bell-shaped and centered at 0 MPa.
- Errors range from roughly -25 MPa to +30 MPa, showing that while the model is unbiased on average, individual prediction swings can be significant.

5. Direct Motivation for Milestone 5 (Model Comparison)

Why We Need Non-Linear Tree-Based Models

Our baseline linear model proves that the features are physically informative ($R^2 \approx 58\%$). However, as demonstrated in our Milestone 2 EDA:

- **Abrams' Law is non-linear:** Compressive strength scales inversely with the water-to-cement ratio.
- **Hydration kinetics are logarithmic:** Strength rises steeply in the first 28 days and flattens later.

In Milestone 5, we will implement non-linear models (Decision Trees, Random Forests, Gradient Boosters) to capture these physical curves and aim to boost R^2 to 80%–90%!